

Product Requirements Document (PRD)

1. Product Overview

Project Title

AeroGuard AI – Personalized Preventive Care System for Tracheostomy Patients

One-Line Pitch

“AeroGuard AI predicts preventable tracheostomy risks before they become serious complications.”

Product Vision

AeroGuard AI is a personalized preventive-care platform designed to identify early warning signs associated with tracheostomy-related infections and airway complications.

Rather than waiting for a complication to become severe, the system compares current patient observations against the patient's personalized baseline and generates a **Prevention Risk Score**.

The platform is intended as a **decision-support and early-warning system**, not a diagnostic or replacement system for doctors.

2. Problem Statement

Tracheostomy patients require continuous monitoring and regular care. Small changes in airway conditions, secretion patterns, stoma appearance, pressure, or care adherence may indicate an increasing risk of complications.

However, routine clinical assessments occur at specific intervals, meaning subtle changes between assessments may not always be identified early.

Existing Challenges

- Difficulty continuously monitoring patients.
- Patient-specific baselines are not always considered.
- Early changes can be difficult to recognize.
- Caregivers may not know when a change requires attention.
- Manual monitoring can be time-consuming.
- Warning signs may become apparent only after the condition has progressed.

Proposed Solution

AeroGuard AI continuously evaluates available patient-care data and compares it with the patient's baseline.

The system produces:

Patient Data → Risk Analysis → Prevention Risk Score → Risk Level → Alert → Recommended Preventive Action

3. Product Goals

Primary Goals

1. Detect potentially concerning changes early.
2. Create a personalized risk score for each patient.
3. Compare current measurements against individual baselines.
4. Identify changes in:
 - Airway-related parameters
 - Pressure
 - Redness
 - Discharge/secretion levels
 - Stoma images
 - Care adherence
5. Provide understandable explanations for the generated risk score.
6. Alert caregivers when risk increases.
7. Maintain patient monitoring history.
8. Provide a simple dashboard for healthcare professionals and caregivers.

Secondary Goals

- Reduce unnecessary manual monitoring.
 - Improve awareness of preventive tracheostomy care.
 - Provide trend visualization.
 - Support earlier communication with healthcare professionals.
-

4. Non-Goals

AeroGuard AI will **not**:

- Diagnose infections.
- Replace doctors or nurses.

- Prescribe medication.
- Automatically make clinical treatment decisions.
- Guarantee that a complication will be prevented.
- Replace emergency medical care.

The system should clearly display that its results are **preventive risk indicators and not medical diagnoses**.

5. Target Users

5.1 Primary Users

Caregivers

Family members or trained caregivers who monitor a patient's condition.

They need:

- Simple risk status.
- Alerts.
- Reasons behind the alert.
- Preventive-care guidance.
- Patient history.

Healthcare Professionals

Doctors and nurses can use the system to:

- Review patient trends.
- View risk scores.
- Examine changes from baseline.
- Review historical observations.
- Support clinical assessment.

5.2 Secondary Users

Patients

Patients may view:

- Current preventive status.
 - Care reminders.
 - Basic preventive guidance.
-

6. Core Features

Feature 1 — Patient Profile

Each patient should have a personalized profile.

Data

- Patient ID
- Name
- Age
- Tracheostomy information
- Baseline pressure
- Baseline redness
- Baseline secretion/discharge
- Baseline care adherence
- Monitoring history

The system should use these values as the patient's personalized baseline.

7. Feature 2 — Patient Data Input

The system should allow caregivers to enter or upload observations.

Input Categories

Category	Example
Pressure	Current airway/tracheostomy pressure
Redness	Percentage/severity value
Discharge	Percentage/severity value
Secretions	Quantity/level
Stoma Image	Uploaded image
Care Adherence	Percentage
Symptoms/Observations	Caregiver-entered observations

Some sensor values can initially be **simulated IoT data** for the prototype.

8. Feature 3 — Personalized Risk Analysis

The system compares current values with the patient's baseline.

Example:

Baseline Pressure = 2.0

Current Pressure = 2.5

Change = 0.5

If the change exceeds the predefined threshold, the risk score increases.

The system can use a **threshold-based decision algorithm** for the initial prototype.

Example Rules

Condition	Risk Points
Pressure changes significantly from baseline	+10
Redness > 40	+15
Discharge > 40	+15
Increased secretion level	+15
Poor care adherence	+10
Concerning image characteristics	+20

The score is capped at:

100

9. Feature 4 — Prevention Risk Score

The system generates a score from:

0–100

Risk Levels

Score	Status	Meaning
0–29	 LOW	No significant changes detected
30–59	 ELEVATED	Changes require closer monitoring
60–100	 HIGH	Significant warning indicators detected

The system should show both the score and the reasons behind it.

Example

Prevention Risk Score: 65/100

Risk Level: HIGH

Changes detected:

- Pressure has changed from baseline.
- Increased redness detected.
- Increased discharge detected.

Recommended action:

Review the patient's condition and contact the appropriate healthcare professional if clinically necessary.

10. Feature 5 — Explainable Risk Analysis

The system should not simply display:

"Risk = 65"

Instead, it should explain **why** the score increased.

Example:

Risk Score: 55

Reasons:

- ✓ Pressure changed from baseline
- ✓ Redness increased
- ✓ Discharge increased

This makes the system easier for caregivers and healthcare professionals to understand.

11. Feature 6 — Stoma Image Analysis

Caregivers can upload an image of the tracheostomy/stoma area.

The computer-vision component can be used to identify visual indicators such as:

- Increased redness
- Visible discharge
- Swelling
- Other visually unusual changes

The result should be treated as a **supporting risk indicator**, not a medical diagnosis.

Example

Image Analysis

Redness indicator: Elevated

Discharge indicator: Moderate

Image-based warning:

Visual changes detected compared with expected appearance.

Please seek professional clinical assessment when appropriate.

12. Feature 7 — Trend Monitoring

The system should store historical measurements.

Example:

Day 1 → Risk 15

Day 2 → Risk 20

Day 3 → Risk 25

Day 4 → Risk 45

Day 5 → Risk 65

A dashboard can display the trend so that users can identify whether risk is:

- Stable
 - Increasing
 - Decreasing
-

13. Feature 8 — Preventive Alerts

When the risk crosses a threshold, the system should generate an alert.

Example

 Elevated Preventive Risk

Patient risk has increased from 25 → 48.

Main contributors:

- Increased redness
- Pressure variation

Action:

Increase monitoring and consider contacting the patient's healthcare professional.

14. Feature 9 — Care Adherence Monitoring

The system can track whether recommended care activities were completed.

Example:

Care Activity	Status
---------------	--------

Stoma care	Completed
------------	-----------

Tube care	Completed
-----------	-----------

Cleaning	Missed
----------	--------

Observation	Completed
-------------	-----------

Low adherence can contribute to the prevention risk score.

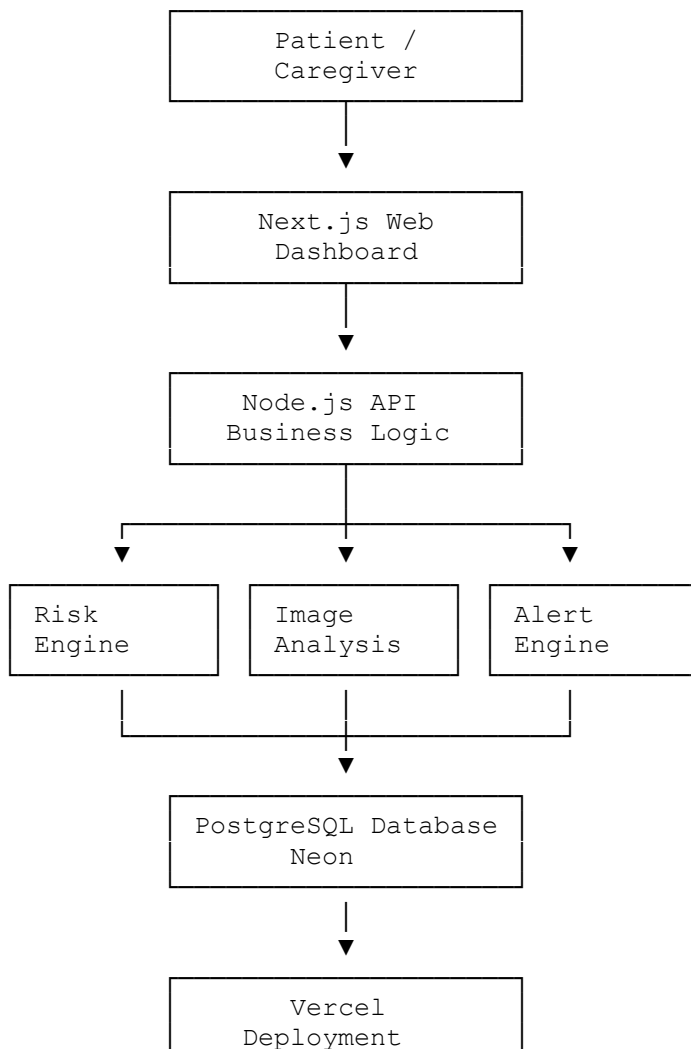
15. Technology Architecture

Proposed Technology Stack

Layer	Technology
Frontend	Next.js
Backend/API	Node.js

Layer	Technology
Database	PostgreSQL
Database Platform	Neon
Hosting	Vercel
AI Assistance	Z.ai
Image Analysis	AI/Computer Vision component
Data Visualization	Next.js dashboard/chart library
Authentication	Application authentication system
Version Control	Git/GitHub

Architecture



16. Database Design

PostgreSQL will store structured application data.

Patients Table

```
patients
-----
id
name
age
baseline_pressure
baseline_redness
baseline_discharge
baseline_adherence
created_at
```

Observations Table

```
observations
-----
id
patient_id
pressure
redness
discharge
secretion_level
care_adherence
image_url
created_at
```

Risk Assessments Table

```
risk_assessments
-----
id
patient_id
risk_score
risk_level
reasons
created_at
```

Alerts Table

```
alerts
-----
id
patient_id
risk_assessment_id
alert_level
message
status
created_at
```

17. Risk Engine

The initial prototype will use a **threshold-based decision algorithm**.

Conceptually:

```
Start
↓
Read patient data
↓
Load patient baseline
↓
Compare current values with baseline
↓
Check predefined thresholds
↓
Add risk points
↓
Calculate total score
↓
Classify risk
↓
Generate reasons
↓
Generate preventive alert
↓
Store assessment
↓
Display result
```

Example

```
Pressure abnormality → +10
Redness > threshold → +15
Discharge > threshold → +15
Adherence problem → +10
```

```
Total = 50
```

```
Risk = ELEVATED
```

18. AI Component

AI can be introduced progressively.

Phase 1 — Rule-Based Risk Engine

Use predefined thresholds and patient-specific baselines.

This makes the prototype:

- Simple
- Explainable
- Easy to test
- Easy to demonstrate during a hackathon

Phase 2 — AI-Assisted Analysis

Z.ai can assist with:

- Generating understandable explanations.
- Summarizing patient trends.
- Producing caregiver-friendly preventive guidance.
- Supporting natural-language interaction with patient data.

Phase 3 — Computer Vision

Image analysis can be introduced to identify visual indicators from stoma images.

Future Phase

With a sufficiently large and clinically validated dataset, a machine-learning model could eventually learn patterns associated with increased risk.

19. Dashboard Requirements

The main dashboard should contain:

Patient Summary

Patient: P001

Current Risk

 65 HIGH

Pressure

 Increased

Redness

 High

Discharge

 Increased

Care Adherence

✓ Good

Risk Trend

Display a graph showing risk over time.

Recent Alerts

Display recent warnings.

Preventive Recommendations

Display simple recommended next steps based on detected changes.

20. User Flow

```
User Login
  ↓
Select Patient
  ↓
View Patient Dashboard
  ↓
Enter / Upload New Observation
  ↓
System Validates Data
  ↓
Risk Engine Processes Data
  ↓
AI/Image Analysis
  ↓
Risk Score Generated
  ↓
Risk Level Assigned
  ↓
Reasons Displayed
  ↓
Preventive Alert Generated
  ↓
Assessment Stored in Database
```

21. Functional Requirements

FR1 — Authentication

Users should be able to securely log in.

FR2 — Patient Management

Authorized users should be able to create and view patient profiles.

FR3 — Baseline Management

The system should store patient-specific baseline values.

FR4 — Data Collection

Users should be able to enter current observations.

FR5 — Image Upload

Users should be able to upload stoma images.

FR6 — Risk Calculation

The system should calculate a 0–100 prevention risk score.

FR7 — Risk Classification

The system should classify patients as Low, Elevated, or High risk.

FR8 — Explanation

The system should display factors contributing to the score.

FR9 — Alerts

The system should generate alerts when predefined risk thresholds are reached.

FR10 — History

The system should store and display historical assessments.

FR11 — Trend Analysis

The system should display changes in risk over time.

FR12 — Preventive Guidance

The system should provide appropriate non-diagnostic preventive guidance.

22. Non-Functional Requirements

Performance

Dashboard requests should respond quickly under normal prototype usage.

Security

Patient information should be protected through:

- Authentication
- Authorization
- Secure database connections
- Environment variables for secrets
- HTTPS
- Restricted access to patient information

Reliability

The risk calculation should produce consistent results for the same inputs.

Scalability

The architecture should allow additional patients, observations, and AI models to be added later.

Usability

The interface should be understandable to users without technical knowledge.

Explainability

Every risk score should provide understandable contributing factors.

23. Privacy and Safety

Because the application deals with healthcare-related information, privacy should be treated as a core requirement.

The prototype should:

- Avoid exposing patient information publicly.
- Use dummy/anonymized patient data for demonstrations.
- Avoid storing unnecessary personal information.

- Secure database credentials.
- Restrict access based on user roles.
- Clearly communicate that the system does not provide a medical diagnosis.

Medical Disclaimer

AeroGuard AI is a preventive-care support and early-warning prototype. It does not diagnose medical conditions, replace healthcare professionals, or provide emergency medical advice.

24. MVP — Minimum Viable Product

For the hackathon, the first working version should focus on the following:

Must Have

- Next.js dashboard
- Patient registration/profile
- PostgreSQL database using Neon
- Patient baseline values
- Current observation input
- Threshold-based risk engine
- 0–100 risk score
- Low/Elevated/High classification
- Explanation of risk factors
- Risk history
- Risk trend chart
- Preventive alerts
- Vercel deployment

Nice to Have

- Stoma image upload
 - Computer-vision analysis
 - Z.ai-powered explanation
 - Care adherence tracking
 - Simulated IoT sensor data
 - Caregiver/doctor roles
-

25. Example Demo Scenario

Initial Patient

Patient ID: P001

Baseline Pressure: 2.0

Baseline Redness: 20

Baseline Discharge: 15

New Observation

Pressure: 2.5

Redness: 55

Discharge: 50

Risk Engine

Pressure changed → +10

Redness > 40 → +15

Discharge > 40 → +15

Total Risk Score = 40

Output



ELEVATED RISK

Risk Score: 40/100

Changes detected:

- Pressure has changed from baseline.
- Increased redness detected.
- Increased discharge detected.

Preventive Status:

Closer monitoring is recommended.

Consider professional clinical assessment if concerning symptoms are present.

26. Success Metrics

The hackathon prototype can be evaluated using:

Metric	Target
Risk calculation accuracy	100% for defined prototype rules
Dashboard response	Fast/near real-time
Risk explanation	Available for every assessment
Patient history	Successfully stored
Alert generation	Triggered at configured thresholds
Data persistence	Successful PostgreSQL storage

Metric	Target
Deployment	Working public/demo deployment
Usability	Simple caregiver workflow

For a real clinical product, additional clinical validation would be required before using the system for patient care.

27. Development Roadmap

Phase 1 — Project Setup

```

Create project
↓
Set up Next.js
↓
Set up Node.js/API structure
↓
Create GitHub repository
↓
Create Neon PostgreSQL database

```

Phase 2 — Database

```

Create database
↓
Create patient table
↓
Create observation table
↓
Create risk assessment table
↓
Create alert table

```

Phase 3 — Backend

```

Create API
↓
Patient APIs
↓
Observation API
↓
Risk calculation API
↓
Alert API

```

Phase 4 — Risk Engine

```

Receive patient data
↓
Compare with baseline
↓
Apply thresholds
↓
Calculate score

```

↓
Assign risk level
↓
Generate explanation

Phase 5 — Frontend

Login
↓
Dashboard
↓
Patient profile
↓
Data entry
↓
Risk score
↓
Alerts
↓
History

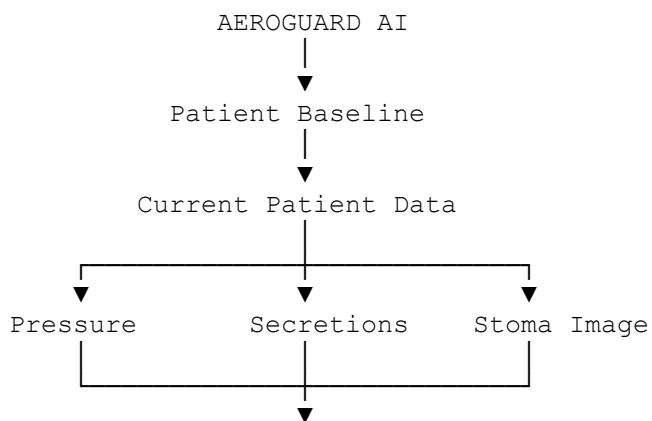
Phase 6 — AI

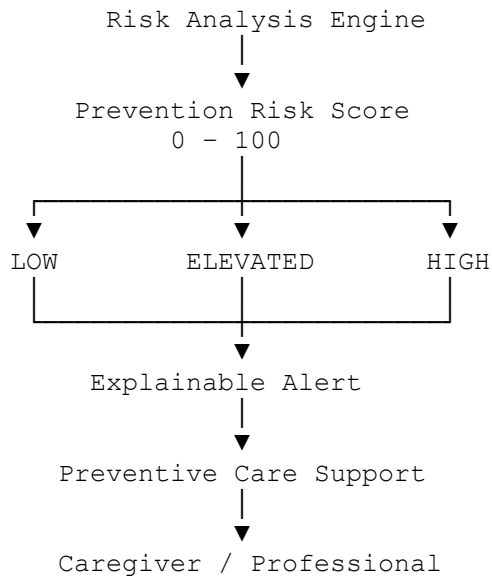
Integrate Z.ai
↓
Generate understandable explanations
↓
Add image analysis
↓
Add AI-assisted preventive guidance

Phase 7 — Deployment

Connect GitHub
↓
Deploy application to Vercel
↓
Connect Neon database
↓
Configure environment variables
↓
Test production application

28. Final Product Flow





29. Expected Outcome

AeroGuard AI will provide a working web-based preventive-care prototype that demonstrates how patient-specific baselines, threshold-based risk analysis, AI assistance, image analysis, and historical monitoring can work together to identify potentially concerning changes earlier.

The final system will demonstrate the concept:

Observe → Compare → Predict Risk → Explain → Alert → Prevent

This approach transforms tracheostomy monitoring from a purely reactive process into a more **personalized and proactive preventive-care workflow**.